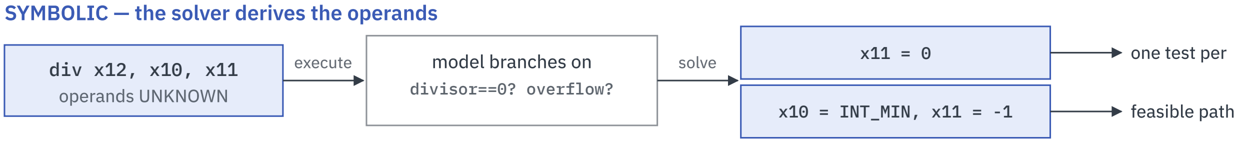
RISC-V SAIL TEST GENERATION · §2.3

ISLA vs Oracle

Two generation backends with opposite limitations, beneath one control loop. Which one serves a target is a routing decision, not a preference — and for two ISA surfaces it is not a decision at all.

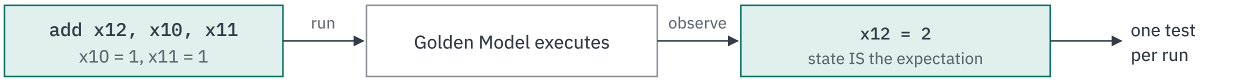
The difference, on one instruction

SYMBOLIC — the solver derives the operands



Aims: derives inputs that provably reach a named behaviour.
Signed overflow is one pair in 2^128 — nothing stumbles on it.

ORACLE — the model reports what happened



Cannot aim — reports what the chosen values did.
But executes anything the model runs, including FP and vector.

No solver, so no representational bound.

The limitations are opposites, which is what makes combining them worthwhile rather than choosing between them. Privileged behaviour needs the machine in a specific state *and* an instruction executed there: the oracle executes anything but cannot construct state; the solver constructs state but cannot execute FP or vector.

Generation backends

ISLA + Z3

Derives operands that provably reach a named behaviour; solves for privileged machine state.

Concrete oracle

Executes anything the model can run, including FP and vector. No solver, so no solver limits.

Templates

Branch and jump only; capped at 4 builders — growth is a defect signal, reported in the suite summary.

Generation routing

Which backend serves a target is a declarative table, not hard-coded dispatch. Most of the ISA surface is reachable by either engine, so routing is a genuine choice; where it is not, the reason is a stated representational limit recorded with the decision.

ISA SURFACE	SYMBOLIC	ORACLE	ROUTED TO
I, M, A, B, C, Zicsr, `Zb*` / `Zc*` / `Za*` / `Zk*`	Yes	Yes	`Symbolic, oracle as fallback`
CSR-only privileged extensions `Sstc`, `Sscofpmf`, state-enable, pointer masking	Where expressible as constraints	Yes	`Either`
F, D, `Zf*` / `Zd*` / `Zh*`	**No** — 45 soft-float primitives absent from the engine	Yes	`Oracle only`
V, `Zv*`, vector crypto	**No** — vector length is runtime state; value representation bounded at 129 bits	Yes	`Oracle only`
Branch, jump	Partial	Partial	`Template — capped at 4`

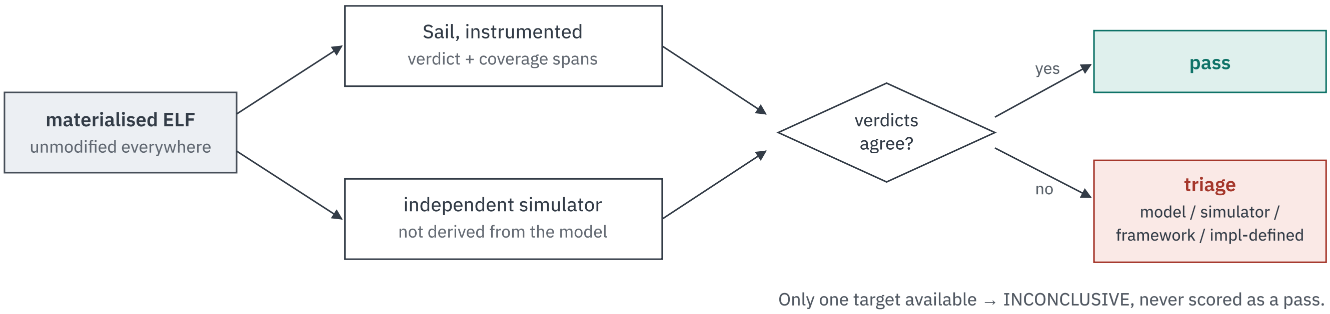
> **The 129-bit bound is a configuration fact, not only an ISA one.** Configurations 5 to 7 of the committed matrix run VLEN 256 and 512, which exceeds the engine's value representation. They are generated and measured by the oracle like any other configuration — a routing consequence, not a coverage gap.

What each target type needs from a generator

TARGET TYPE	WHAT IT COVERS	BEST SUITED FOR
Instruction sequences	ALU ops, loads and stores	Functional execution and corner cases
CSRs / state	`Zicsr`, counters, state enable, privilege state	CSR access, privilege transitions, machine-state scenarios
Traps / exceptions	Illegal instruction, page faults, `ecall` / `ebreak`	Exception handling, trap entry and return, fault scenarios
Interrupts	Timer, external, software	Delivery, priority, delegation
Translation / memory	Page tables, Sv modes, PMP/PMA	Virtual memory, access faults, protection
Privileged features	M/S/U modes, counters, state-enable, QoS	Mode transitions, delegation, feature enablement

Routing is stated once, in the table above — this one says what the work *is*, not which engine does it.

What the framework does with the result



The independent simulator is the only component not derived from the Sail model, which is what stops a suite that agrees with itself from being mistaken for a suite that is correct.

The rest of the pipeline

What counts as 100% coverage, how the privileged architecture becomes measurable targets, the committed scope and the seven configurations are on the companion page — *Sail Model to Coverage Report*.

Content traces to the technical plan, sections 1.2, 1.4, 2.3 and 2.5. Routing is stated once, in Generation routing.